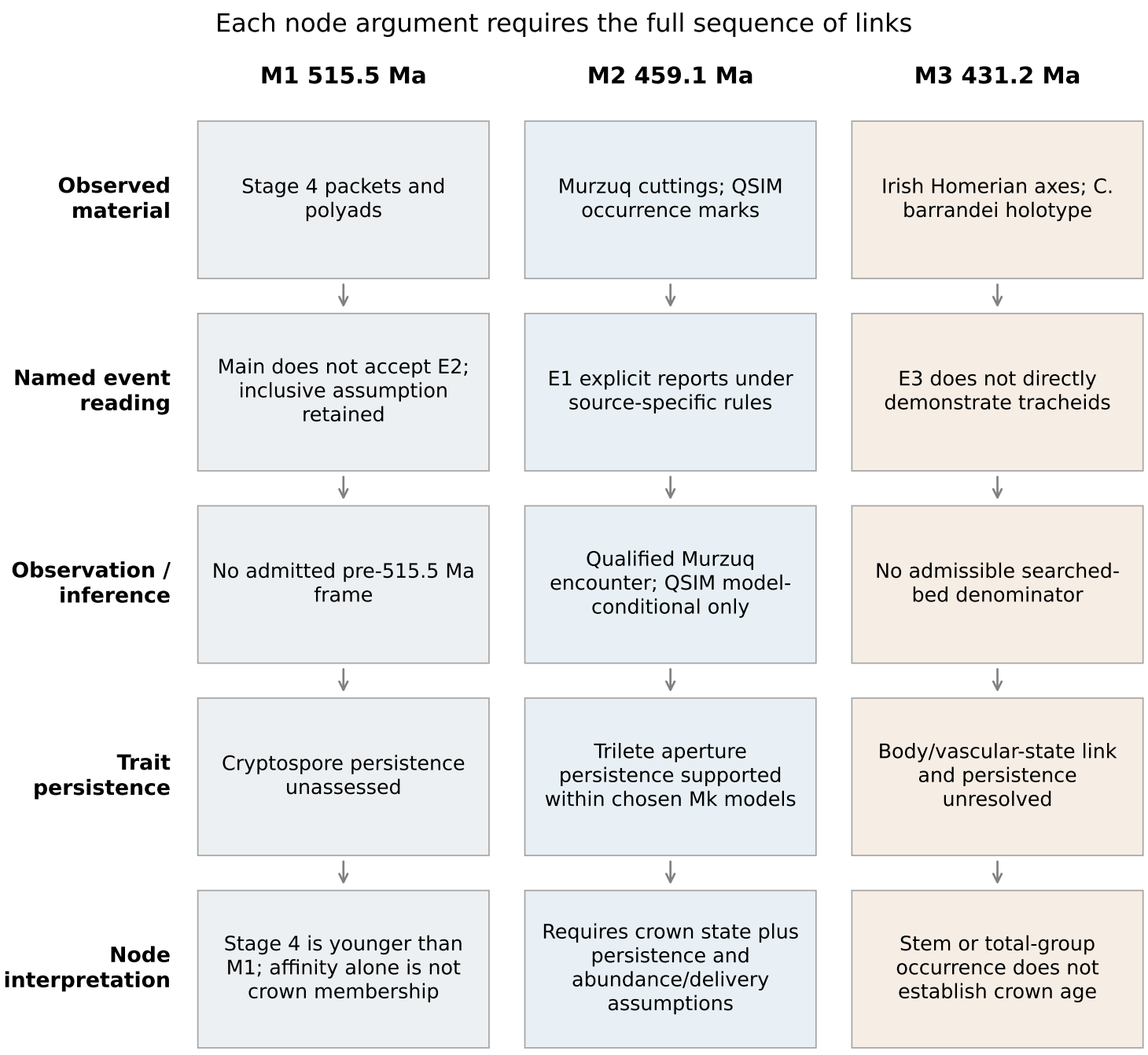
Each node argument requires the full sequence of links
M1 515.5 Ma
M2 459.1 Ma
M3 431.2 Ma
Observed material
Stage 4 packets and polyads
Murzuq cuttings; QSIM occurrence marks
Irish Homerian axes; C. barrandei holotype
Named event reading
Main does not accept E2; inclusive assumption retained
E1 explicit reports under source-specific rules
E3 does not directly demonstrate tracheids
Observation / inference
No admitted pre-515.5 Ma frame
Qualified Murzuq encounter; QSIM model-conditional only
No admissible searched-bed denominator
Trait persistence
Cryptospore persistence unassessed
Trilete aperture persistence supported within chosen Mk models
Body/vascular-state link and persistence unresolved
Node interpretation
Stage 4 is younger than M1; affinity alone is not crown membership
Requires crown state plus persistence and abundance/delivery assumptions
Stem or total-group occurrence does not establish crown age